BASE CHANGE OF IMMERSIONS: SURJECTIVITY OF THE CONORMAL COMPARISON MAP AND ITS FLAT CASE ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 556-570 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 572-588, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The existence of etale surjective presentations (Spaces, lift-morphism-presentations), the etale localization of the conormal sheaf and the functoriality map (same chapter), and the schemes case (Morphisms, conormal-functorial-flat); cited inputs with the descent reduction written in the retained span.

Source reference. Original lemma 04CP: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 455–471. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04G2: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 520–548. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04CO: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 432–441. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0473: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 588–588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02X1: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 588–588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let*

$$\begin{array}{ccc} Z & \xrightarrow{i} & X \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ Z' & \xrightarrow{i'} & X' \end{array}$$

be a fibre product diagram of algebraic spaces over S. Assume i, i' immersions. Then the canonical map $f^*\mathcal{C}_{Z'/X'} \to \mathcal{C}_{Z/X}$ of Lemma 04CP is surjective. If g is flat, then it is an isomorphism.

Proof. Choose a commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & X \\ \downarrow & & \downarrow \\ U' & \longrightarrow & X' \end{array}$$

where U, U' are schemes and the horizontal arrows are surjective and étale, see Spaces, Lemma 02X1. Then using Lemmas 04CO and 04G2 we see that the question reduces to the case of a morphism of schemes. In the schemes case this is Morphisms, Lemma 0473. □